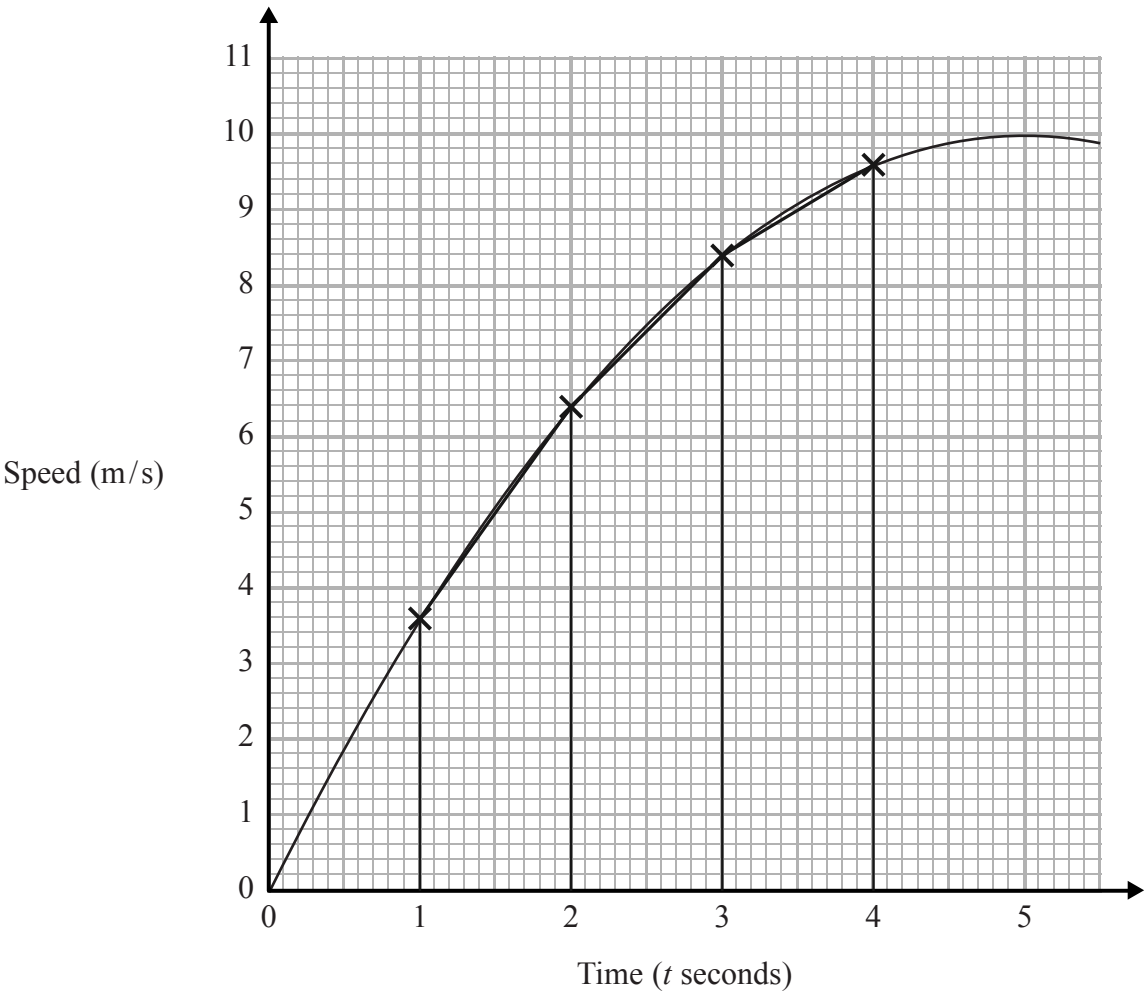
17 Here is a speed-time graph showing the speed, in metres per second, of an object t seconds after it started to move.



(a) Use 3 strips of equal width to find an estimate for the area under the graph between $t = 1$ and $t = 4$

At $t = 1, 2, 3, 4$: speeds $\approx 3.6, 6.4, 8.4, 9.6$.

$A_1 = \frac{1}{2}(3.6 + 6.4) \times 1 = 5$

$A_2 = \frac{1}{2}(6.4 + 8.4) \times 1 = 7.4$

$A_3 = \frac{1}{2}(8.4 + 9.6) \times 1 = 9$

Total $= 5 + 7.4 + 9 = 21.4$

1 mark
Read end heights 3.6 and 9.6

1 mark
$\frac{1}{2}(a + b)h$ for each trapezium

1 mark
21.4

(3)

(b) Describe fully what your answer to part (a) represents.

Estimated distance travelled from 1 s to 4 s,

about 21.4 metres.

1 mark
Distance estimate

1 mark
Metres

(2)

(c) Explain whether your answer in part (a) gives an underestimate or an overestimate for the area under the graph.

Underestimate: the straight tops of the trapezia

lie below the curve.

1 mark
Chords below curve

(1)

(Total for Question 17 is 6 marks)